Area of Trapeziums

Name _____ Date _____ Calculator allowed Version 1

Scan to practise this topic online

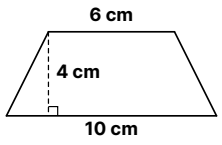
KEY WORDS parallel sides · perpendicular height *(teitei hāngai)* · average *(toharite)* · expression *(kīanga taurangi)*

A The formula ●○○

Remember: Area = $\frac{1}{2}$ × (sum of the parallel sides) × height. Add the two parallel sides first, then halve, then multiply by the height.

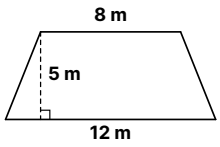
1. Find the area of each trapezium.

a)



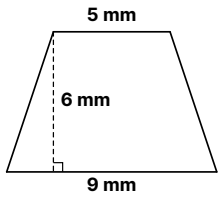
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b)



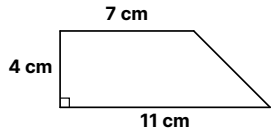
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c)



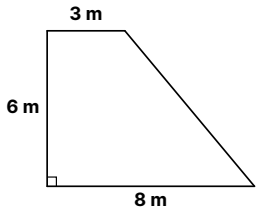
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d)



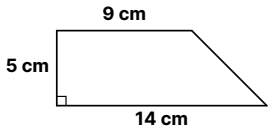
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e)



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f)



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B Finding a missing length ●●○

Remember: Put what you know into the formula and solve step by step: undo the $\frac{1}{2}$ and the height first, then the addition.

2. Find the missing length.

a) A trapezium has parallel sides of 9 cm and 7 cm and an area of 40 cm². Find its height.

b) A trapezium has an area of 54 m², a height of 6 m and one parallel side of 8 m. Find the other parallel side.

c) A trapezium has parallel sides of 14 cm and 8 cm and an area of 66 cm². Find its height.

d) A trapezium has an area of 45 mm², a height of 5 mm and one parallel side of 11 mm. Find the other parallel side.

e) A trapezium has parallel sides of 8 m and 6 m and an area of 35 m². Find its height.

f) A trapezium has an area of 72 cm², a height of 8 cm and one parallel side of 10 cm. Find the other parallel side.

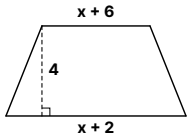
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C Lengths with algebra ●●●

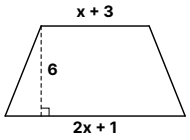
Remember: Add the two parallel sides as expressions first, then halve and multiply by the height.

3. Write an expression for the area of each trapezium. Simplify it.

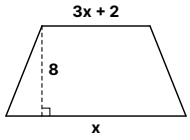
a)



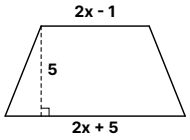
b)



c)

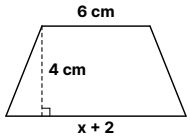


d)



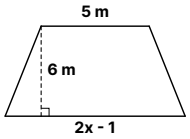
4. The area of each trapezium is given. Write an equation and solve it to find x.

a)



The area is 26 cm². Find x.

b)



The area is 36 m². Find x.

D Problems ●●●

5. The cross-section of a canal is a trapezium: 12 m wide at the top, 4 m wide at the bottom and 3 m deep. The canal is 40 m long. Find the volume of water it holds when full.

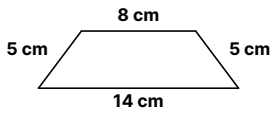
6. A roof panel is a trapezium with parallel sides of 9 m and 5 m, 2.5 m apart. There are 2 such panels. Find the total area of the panels.

7. A garden bed is a trapezium with parallel sides of 7 m and 4 m and a width of 3 m between them. Compost costs $18 per square metre. Find the cost of compost for the bed.

8. Explain why the trapezium formula works, by cutting a trapezium with parallel sides a and b and height h into a rectangle and two triangles, or into two triangles along a diagonal.

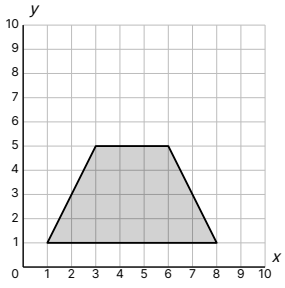
★ Extension

9. An isosceles trapezium has parallel sides of 14 cm and 8 cm, and slant sides of 5 cm. Find its height first, then its area.



10. A trapezium has corners at (1, 1), (8, 1), (6, 5) and (3, 5).

a) Find its area in square units.



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Area of Trapeziums

Name _____ Date _____ Calculator allowed Version 2

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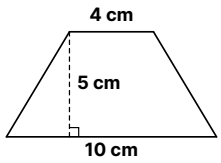
KEY WORDS parallel sides · perpendicular height *(teitei hāngai)* · average *(toharite)* · expression *(kīanga taurangi)*

A The formula ●○○

Remember: Area = $\frac{1}{2}$ × (sum of the parallel sides) × height. Add the two parallel sides first, then halve, then multiply by the height.

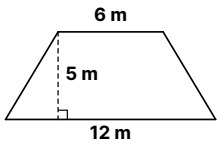
1. Find the area of each trapezium.

a)



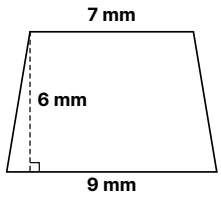
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b)



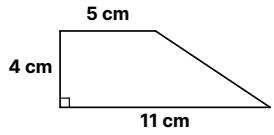
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c)



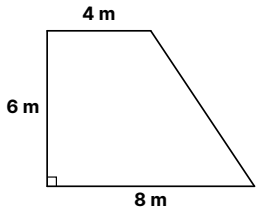
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d)



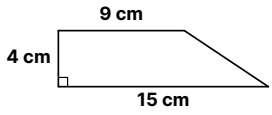
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e)



_____

f)



_____

B Finding a missing length ●●○

Remember: Put what you know into the formula and solve step by step: undo the $\frac{1}{2}$ and the height first, then the addition.

2. Find the missing length.

a) A trapezium has parallel sides of 8 cm and 4 cm and an area of 36 cm². Find its height.

b) A trapezium has an area of 60 m², a height of 6 m and one parallel side of 9 m. Find the other parallel side.

c) A trapezium has parallel sides of 13 cm and 9 cm and an area of 77 cm². Find its height.

d) A trapezium has an area of 50 mm², a height of 5 mm and one parallel side of 12 mm. Find the other parallel side.

e) A trapezium has parallel sides of 9 m and 5 m and an area of 42 m². Find its height.

f) A trapezium has an area of 84 cm², a height of 8 cm and one parallel side of 11 cm. Find the other parallel side.

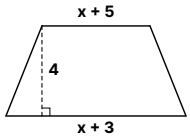
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C Lengths with algebra ●●●

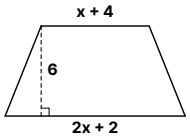
Remember: Add the two parallel sides as expressions first, then halve and multiply by the height.

3. Write an expression for the area of each trapezium. Simplify it.

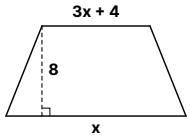
a)



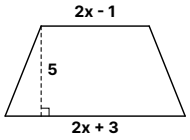
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c)

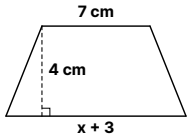


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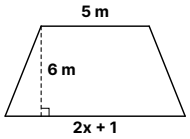
4. The area of each trapezium is given. Write an equation and solve it to find x.

a)



The area is 32 cm². Find x.

b)



The area is 36 m². Find x.

D Problems ●●●

5. The cross-section of a canal is a trapezium: 10 m wide at the top, 4 m wide at the bottom and 3 m deep. The canal is 50 m long. Find the volume of water it holds when full.

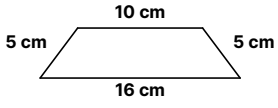
6. A roof panel is a trapezium with parallel sides of 8 m and 4 m, 3 m apart. There are 2 such panels. Find the total area of the panels.

7. A garden bed is a trapezium with parallel sides of 8 m and 5 m and a width of 3 m between them. Compost costs $16 per square metre. Find the cost of compost for the bed.

8. Explain why the trapezium formula works, by cutting a trapezium with parallel sides a and b and height h into a rectangle and two triangles, or into two triangles along a diagonal.

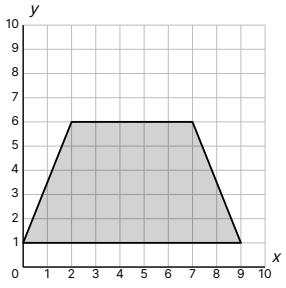
★ Extension

9. An isosceles trapezium has parallel sides of 16 cm and 10 cm, and slant sides of 5 cm. Find its height first, then its area.



10. A trapezium has corners at (0, 1), (9, 1), (7, 6) and (2, 6).

a) Find its area in square units.



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